

Baby Steps: A Staged Validation Protocol

Evaluating an Embodied, Delayed-Reveal Pedagogy for Adult Beginner and Re-Entering Piano Learners

Protocol v1.0 | June 2026

Document type. Pre-registration-ready study protocol describing a four-phase validation program for the Baby Steps instructional framework. The protocol specifies aims, hypotheses, design, eligibility, interventions, outcomes, sample size, procedures, analysis plan, ethics, accessibility, and reporting commitments. No results are reported. Submission to an independent ethics review and to a public protocol registry is planned prior to Phase 3.

Structured Abstract

Background. Adult novice and re-entering pianists studying outside traditional studio settings are underserved by default pedagogical sequencing. Baby Steps is an evidence-informed instructional framework that defines the smallest unit of instruction as a small embodied playable action, postpones the introduction of symbolic theory language until after perceptual and motor contact, and operationalizes progression through three observable states (Practiced, Confident, Mastered).

Aim. To evaluate, in a four-phase staged program, the content validity, usability, accessibility, short-term learning effects, and comparative effects of the Baby Steps framework relative to an explanation-first comparator.

Methods. Phase 1 is an expert content review (6–10 piano pedagogues, 2–4 adult-learning specialists) using a structured rubric. Phase 2 is a moderated usability and accessibility study ($n = 12\text{--}20$). Phase 3 is a single-arm pilot learning study ($n = 30\text{--}50$) over 6–8 weeks with pre/post and delayed measures. Phase 4 is a two-arm parallel-group comparative study ($n = 60\text{--}120$) with 1:1 randomization to experience-first or explanation-first sequencing.

Primary outcomes. Phase 3: change from baseline on a standardized left-hand harmonic motion accuracy battery at 6–8 weeks. Phase 4: between-group difference at 6–8 weeks on the same battery, with co-primary endpoints of delayed recognition of named concepts (24–72 hours and 1–2 weeks) and near transfer to a structurally related novel pattern.

Status. Phases 1 and 2 are scheduled. Phase 3 will be pre-registered prior to recruitment. Phase 4 will be pre-registered separately after Phase 3 results.

1. Background and Rationale

Conventional piano instruction often begins with external structures — notation, terminology, technique rules, and theoretical definitions — that, presented simultaneously to a novice, can exceed working-memory capacity (Sweller, van Merriënboer, & Paas, 2019) and produce avoidance and dropout,

especially for adult learners with attentional variability (Beheshti, Chavanon, & Christiansen, 2020). Within music learning, scaffolded practice that integrates modeling, recorded playback, and self-evaluation produces larger gains than unscaffolded practice (Hewitt, 2001; Rom, 2024), self-regulation instruction improves perceived practice and self-efficacy (Miksza, 2015), and an effects-of-movement attention focus — including focus on sound — supports more accurate and generalizable motor performance at the keyboard (Duke, Cash, & Allen, 2011).

The Baby Steps framework integrates these findings into a single instructional model designed for adult novice and re-entering pianists. Its central pedagogical claim is that early piano learning is improved when each unit of instruction is sequenced from a small embodied action through perceptual noticing, symbolic naming, formal theory, and transfer to musical use. Because direct empirical support for this sequence in the named population is currently absent, the present protocol specifies how the framework will be evaluated.

2. Objectives and Hypotheses

2.1 Aims by phase

Phase	Aim
1. Expert content review	Establish content validity of the framework, the curriculum scope, the practice-state rubric, and the claim ladder.
2. Usability and accessibility	Identify friction, overload, and accessibility failures across representative learner profiles and configurations.
3. Pilot learning study	Estimate short-term gains in left-hand harmonic motion accuracy, delayed recognition of named concepts, near transfer, perceived overwhelm, and self-efficacy, in the target population using the framework.
4. Comparative study	Test experience-first (Touch–Move–Feel–Notice–Name) sequencing against an explanation-first comparator on the same outcomes.

2.2 Phase 4 primary hypotheses

ID	Hypothesis
H1	At 6–8 weeks, participants in the experience-first arm will show non-inferior or superior accuracy on the standardized left-hand harmonic motion battery relative to the explanation-first arm.
H2	At 24–72 hours and 1–2 weeks post-introduction, the experience-first arm will show equal or better delayed recognition of named harmonic concepts.
H3	The experience-first arm will show equal or better near transfer (same harmonic motion in a new key) at 6–8 weeks.
H4	The experience-first arm will report lower perceived overwhelm and higher self-efficacy across the study window.

ID	Hypothesis
H5	The experience-first arm will show higher 7-day and 30-day return-after-lapse rates.

3. Methods

3.1 Trial design

The validation is structured as a staged program of four phases. Phases 1 and 2 are qualitative and mixed-methods; Phase 3 is a single-arm pilot; Phase 4 is a parallel-group, two-arm, individually randomized controlled study with 1:1 allocation. Each phase is gated by the prior phase: results that fail predefined go/no-go criteria trigger framework revision before progression.

3.2 Setting

All phases are conducted remotely. Participants use their own keyboard or piano (61–88 keys; MIDI strongly preferred but not required) and a personal device (tablet, laptop, or phone) running the Baby Steps app. Phase 2 sessions are moderated by video conference. Phases 3 and 4 are unmoderated; assessments are delivered in-app and via short web surveys.

3.3 Eligibility criteria

Category	Inclusion	Exclusion
Age	≥ 18 years.	< 18 years.
Experience	Self-reported novice (< 6 months cumulative lifetime study) or re-entering learner (no regular practice in the past 24 months).	Currently studying privately with a piano teacher; conservatory students; professional or semi-professional pianists.
Instrument access	Daily access to a keyboard or piano of ≥ 61 keys for the duration of the study.	No reliable instrument access.
Device	Personal smartphone, tablet, or computer capable of running the app.	No personal device.
Language	Fluent in the study language (initially English).	Insufficient language fluency to complete consent and instructions.
Health	No condition that the participant indicates would interfere with self-directed practice.	Participant indicates a condition that would interfere; we do not screen for or collect medical diagnoses.

3.4 Interventions

Experimental intervention (all phases). The Baby Steps framework as defined in the v1.0 white paper, instantiated as the *Left-Hand Chord Freedom* path. Each Baby Step satisfies the six operational criteria (single new element; playable on first encounter; audible meaning; reversible; theory-deferred; transferable). Progression is governed by the Practiced / Confident / Mastered rubric.

Comparator (Phase 4 only). An explanation-first sequencing of the same content. The comparator presents the theoretical label and a written explanation before the first playable attempt, then proceeds to the same playable material and the same transfer destinations. Total time-on-task and total exposure to the same harmonic patterns are matched across arms; the only manipulated factor is the order in which the label and the action are introduced within each unit.

3.5 Outcomes

Outcome	Instrument	Timing	Phase
Left-hand harmonic motion accuracy	Standardized 12-pattern MIDI-scored battery (pitches, order, tempo stability).	Baseline; week 4; week 6–8.	3, 4
Delayed recognition of named concept	Forced-choice + free-recall short quiz on previously introduced labels.	24–72 hours and 1–2 weeks post-introduction.	3, 4
Near transfer	Same harmonic motion in a new key; MIDI-scored.	Week 6–8.	3, 4
Perceived overwhelm	Single-item 0–10 NRS after each session.	Every session.	2, 3, 4
Self-efficacy	Short adapted music self-efficacy scale.	Baseline; week 4; week 6–8.	3, 4
Return-after-lapse rate	Telemetry: % learners returning within 7 and 30 days of a ≥ 5 -day lapse.	Continuous.	3, 4
Quit-on-error rate	Telemetry: session-ends within 30s of an error event.	Continuous.	2, 3, 4
Accessibility parity	Completion rate and time-on-task across configurations (reduced motion; high contrast; on-screen keyboard).	Throughout.	2, 3, 4
Safety / harms	Open-text report of physical discomfort, frustration events, distress.	Every session prompt; debrief.	All

3.6 Sample size and stopping considerations

Phase 1. Saturation-based; 6–10 pedagogues plus 2–4 adult-learning specialists are expected to be sufficient for content-validity stability.

Phase 2. 12–20 participants distributed across representative profiles and accessibility configurations; usability findings typically saturate by $n = 12$.

Phase 3. 30–50 participants provide adequate precision for pilot estimates of within-group change and for refining the Phase 4 effect-size assumption. No formal hypothesis test is planned; analysis is descriptive with 95% confidence intervals.

Phase 4. 60–120 participants total. The final n will be determined after Phase 3 using the observed standard deviation and an adjustable minimal clinically meaningful difference. A planned interim analysis at 50% accrual will support a futility stop if the conditional power falls below 20%.

3.7 Recruitment

Recruitment will use the existing Baby Steps interest list, paid social ads in adult-learner communities, and outreach to adult-learner forums. Screening is self-report and conducted in a brief web form. All paid ads will include the same neutral framing of the study (no efficacy claim) approved by the ethics review.

3.8 Allocation and blinding (Phase 4)

Allocation is 1:1 to experience-first or explanation-first using a computer-generated stratified randomization (strata: self-reported attentional concerns yes/no; re-entering vs. true novice). Participants are not blinded to their condition because the difference is experiential. Outcome scoring is automated for MIDI-scored measures and therefore blinded by design; survey scoring is automated; human review of free-text harms reports is performed by reviewers blinded to arm.

3.9 Procedures

Phase	Key procedure steps
1	(a) Reviewers receive the framework, curriculum map, rubric, and three lesson exemplars. (b) Reviewers complete a structured rubric on clarity, sequencing, scope, claim language, and accessibility. (c) Two debrief calls. (d) Framework revision.
2	(a) Onboarding. (b) Moderated 60-minute session per participant: think-aloud through three Baby Steps and one mastery checkpoint. (c) Post-session questionnaire. (d) Accessibility audit pass across reduced motion, high contrast, and on-screen keyboard configurations.
3	(a) Consent and baseline. (b) 6–8 weeks of unmoderated app use, ≥ 3 sessions/week target. (c) Mid-point and end-point assessments. (d) Delayed-recognition probes at scheduled intervals. (e) Optional 30-minute debrief interview with a stratified subsample.
4	(a) Consent and baseline. (b) Randomization. (c) Six to eight weeks of arm-specific delivery. (d) Mid-point and end-point assessments. (e) Delayed-recognition and transfer assessments. (f) Pre-specified primary and secondary analyses.

3.10 Data collection and management

Telemetry includes per-attempt MIDI captures, accuracy scores, tempo, abort events, session timings, and progress-state transitions. Surveys are delivered in-app and scored automatically. All data are stored under participant-specific pseudonymous identifiers; linkage to identifying information is held separately by the study coordinator. The data dictionary, including all variable names, definitions, and scoring rules, is finalized before the start of each phase and published with the pre-registration.

3.11 Statistical analysis plan

Phase 3. Descriptive statistics with 95% confidence intervals for each outcome; within-participant change from baseline with paired comparisons; survival-style analysis of session adherence and return-after-lapse.

Phase 4. Intention-to-treat analysis. Primary endpoint analyzed with mixed-effects models adjusting for baseline accuracy and stratification factors. Co-primary endpoints (delayed recognition; near transfer) use the same modeling approach. Multiplicity is controlled via a hierarchical testing procedure across H1–H5. Secondary outcomes are analyzed using the same models without multiplicity correction and are reported as exploratory. Missing data are handled by multiple imputation under a missing-at-random assumption, with sensitivity analyses under plausible deviations.

3.12 Adverse-event and harms monitoring

Although the intervention is educational, the protocol commits to monitoring for physical discomfort (e.g., hand pain), distress, and significant frustration. Each session prompts a brief check; any report exceeding a pre-specified threshold triggers a coach message, an offer to stop, and a study-team review. Serious adverse events are logged and reviewed within 72 hours. No clinical assessment is performed; participants are directed to appropriate external resources when warranted.

4. Ethics, Consent, and Oversight

The protocol will be submitted to an independent ethics review board prior to Phase 3. Participants provide informed digital consent that explicitly states: the study is educational, not clinical; no diagnosis or treatment is offered; participation is voluntary; participants can withdraw at any time without affecting their access to the product; and what data are collected and how they are stored.

Compensation is modest and proportionate to participant time. Participants in Phases 3 and 4 retain access to the framework after the study window. The study team includes at least one investigator without a direct commercial interest in the framework's adoption; that investigator chairs the analysis plan review and any decision about deviation from the pre-registration.

5. Accessibility and Inclusion

The protocol commits to specific accessibility behaviors following the structure of WCAG 2.2 (W3C, 2023). Phase 2 explicitly audits perception, operability, understandability, and robustness. Phases 3 and 4 monitor completion parity across reduced motion, high-contrast, on-screen-keyboard, captions-on, and slow-tempo configurations. Recruitment materials are reviewed for plain-language readability. ADHD-aware design choices are framed as instructional and UX patterns, not as clinical interventions, and no diagnostic data are collected.

6. Reporting, Pre-registration, and Data Sharing

Phases 3 and 4 will be pre-registered prior to recruitment in a public registry. Reporting will follow CONSORT for Phase 4. The full analysis code and a de-identified dataset will be released after publication, subject to participant consent and applicable privacy law. Any deviation from the pre-registration will be transparently disclosed in the report.

Claim language in product-facing communications will be aligned with the most advanced phase completed. Until Phase 1 is complete, the framework is described as 'evidence-informed.' After Phase 1, the framework adds 'reviewed by independent piano pedagogues.' After Phase 3, the product may

describe specific learning gains observed in the pilot, with sample size and limitations. Only after Phase 4 may comparative-superiority language be used, bounded to the tested population and outcomes.

7. Limitations

The protocol is constrained by several factors. First, blinding of participants is not possible because the intervention is experiential. Second, recruitment via the framework's existing interest list risks selection toward enthusiastic adopters; this is partially mitigated by paid recruitment in independent communities. Third, the primary outcome battery is specific to left-hand harmonic motion and does not test broader pianistic competence; generalization beyond this domain is out of scope. Fourth, the population is drawn from English-speaking learners with self-provided keyboards and devices, limiting cultural and economic generalizability. Fifth, the protocol does not test long-term retention beyond a few weeks; a long-term retention extension is planned separately.

8. Timeline

Milestone	Target window
Phase 1 expert content review	Months 1–2
Framework revision	Month 3
Phase 2 usability and accessibility	Months 3–4
Phase 3 pre-registration	Month 4
Phase 3 pilot learning study	Months 5–7
Phase 3 analysis and report	Month 8
Phase 4 pre-registration and ethics submission	Months 8–9
Phase 4 comparative study	Months 10–13
Phase 4 analysis and report	Months 14–15

References

- Beheshti, A., Chavanon, M. L., & Christiansen, H. (2020). Emotion dysregulation in adults with attention deficit hyperactivity disorder: A meta-analysis. *BMC Psychiatry*, 20(1), 120.
- Chan, A. W., Tetzlaff, J. M., Altman, D. G., et al. (2013). SPIRIT 2013 statement: Defining standard protocol items for clinical trials. *Annals of Internal Medicine*, 158(3), 200–207.
- Duke, R. A., Cash, C. D., & Allen, S. E. (2011). Focus of attention affects performance of motor skills in music. *Journal of Research in Music Education*, 59(1), 44–55.
- Hewitt, M. P. (2001). The effects of modeling, self-evaluation, and self-listening on junior high instrumentalists' music performance and practice attitude. *Journal of Research in Music Education*, 49(4), 307–322.
- Leman, M. (2010). An embodied approach to music semantics. *Musicae Scientiae*, 14(1S), 43–67.
- McPherson, G. E., Osborne, M. S., Evans, P., & Miksza, P. (2019). Applying self-regulated learning microanalysis to study musicians' practice. *Psychology of Music*, 47(1), 18–32.
- Miksza, P. (2015). The effect of self-regulation instruction on the performance achievement, musical self-efficacy, and practicing of advanced wind players. *Psychology of Music*, 43(2), 219–243.
- Roediger, H. L., & Karpicke, J. D. (2006). Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science*, 17(3), 249–255.
- Rom, B. (2024). The effects of scaffolded practice conditions on string players' performance accuracy and practice efficiency. *Journal of Research in Music Education*. (Advance online publication.)
- Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68–78.
- Schulz, K. F., Altman, D. G., Moher, D., & CONSORT Group. (2010). CONSORT 2010 statement: Updated guidelines for reporting parallel group randomized trials. *BMJ*, 340, c332.
- Substance Abuse and Mental Health Services Administration. (2014). *SAMHSA's concept of trauma and guidance for a trauma-informed approach* (HHS Publication No. SMA 14-4884).
- Sweller, J., van Merriënboer, J. J. G., & Paas, F. (2019). Cognitive architecture and instructional design: 20 years later. *Educational Psychology Review*, 31(2), 261–292.
- W3C. (2023). *Web Content Accessibility Guidelines (WCAG) 2.2*.